

Project Proposal in CS 118 - Thesis Writing 1

Project Title: NutriLink: AI-Driven Nutrition Intervention and Feeding Management Ecosystem

Project Objectives:

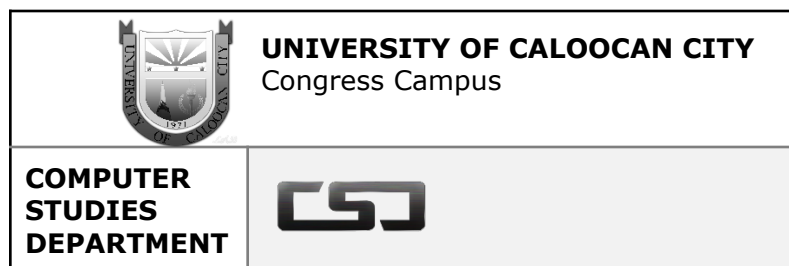
General: To develop an IoT-based nutrition assessment and feeding management ecosystem titled NutriLink that automates nutritional screening, provides AI-driven nutrition recommendations and intelligent food allocation, and utilizes blockchain technology to ensure secure, transparent, and traceable management of nutrition records, feeding transactions, and inventory operations through an integrated mobile application, Smart Nutrition Station, and web-based dashboard.

Specific:

1. To design and build an IoT-based Smart Nutrition Station that measures height, weight, body temperature, and Body Mass Index (BMI) for nutritional assessment.
2. To develop a mobile application that enables user registration, QR Code identification, nutrition monitoring, and feeding history access.
3. To develop an AI-based predictive model that evaluates nutritional status and recommends appropriate nutrition interventions.
4. To implement an intelligent nutrition allocation and automated dispensing system based on AI recommendations and beneficiary eligibility.
5. To implement an AI-based forecasting model that predicts food demand, inventory consumption, shortages, and potential food waste.
6. To integrate a blockchain-based ledger that records nutrition assessments, feeding transactions, inventory movement, and intervention history for transparency and traceability.
7. To develop a real-time web dashboard that monitors nutrition records, inventory status, AI predictions, and blockchain transactions.
8. To generate analytical reports that support nutrition monitoring and resource management.

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Revised: 11, 2022



Scope:

1. The system will be implemented as a prototype consisting of an IoT-based Smart Nutrition Station, a mobile application, and a centralized web dashboard.
2. The project covers user registration, nutritional assessment, AI-based nutrition analysis, automated nutrition dispensing, inventory monitoring, and blockchain transaction recording.
3. The system provides real-time monitoring, historical records, and analytics to support nutrition intervention and feeding management.

Limitations:

1. The prototype is limited to basic nutritional assessment using height, weight, body temperature, and Body Mass Index (BMI).
2. AI-generated recommendations are intended to assist nutrition monitoring and should not replace professional medical evaluation.
3. The blockchain implementation uses a permissioned blockchain network for prototype and educational purposes.
4. The automated dispensing mechanism is limited to pre-packaged nutrition packs or meal vouchers.

Features:

1. IoT-Based Smart Nutrition Station
2. Automatic Height, Weight, Temperature, and BMI Measurement
3. Mobile Application with QR Code-Based User Identification
4. AI-Based Nutritional Assessment and Recommendation
5. Intelligent Nutrition Allocation and Automated Dispensing
6. AI-Based Food Demand and Inventory Forecasting
7. Blockchain-Based Nutrition and Inventory Records
8. Real-Time Web Dashboard and Monitoring
9. Nutrition Progress Tracking and Historical Records
10. Automated Reports and Data Analytics

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